

Neuro–Symbolic Interpretation of the Cl(3, 3) Electromagnetic–Sphere Verification Suite

Appendix F machine–verified identities — a cold reading of the literal Wolfram output

Electromagnetic–Sphere programme • Part I foundations
Prepared by a deterministic neuro–symbolic interpreter • all claims anchored to kernel output

Provenance and method. The nine source files (`representation.wl`, 01–07, `run_all.wl`) were executed in a Wolfram Language kernel. Because the kernel is stateless, the algebra of `representation.wl` was inlined verbatim into each evaluation rather than loaded by path. The real 8×8 generators are the Jordan–Wigner Kronecker products

$$e_1 = \sigma_x \otimes \mathbf{1} \otimes \mathbf{1}, \quad e_2 = \sigma_z \otimes \sigma_x \otimes \mathbf{1}, \quad e_3 = \sigma_z \otimes \sigma_z \otimes \sigma_x, \quad e_4 = \varepsilon \otimes \mathbf{1} \otimes \mathbf{1}, \quad e_5 = \sigma_z \otimes \varepsilon \otimes \mathbf{1}, \quad e_6 = \sigma_z \otimes \sigma_z \otimes \varepsilon,$$

with $\varepsilon = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ the real “ i ” ($\varepsilon^2 = -\mathbf{1}$) and metric $\eta = \text{diag}(+, +, +, -, -, -)$. Equality tests use `Simplify[...]==0`; spectra use `Eigenvalues`; the gravity contraction uses seeds 171 ($n = 4$) and 66 ($n = 6$). The only kernel messages were `Symbol::undefined2` and `Solve::svars`, both benign (forward references and an under-determined linear system whose free direction is the intended result). Every numeric and Boolean value quoted in Section 1 is the literal value the kernel printed.

Status legend. **[EXACT]** proven exactly by symbolic computation; **[DERIVED]** a physical reading rigorously implied by an **[EXACT]** identity plus a stated convention; **[AXIOM]** assumed as input/definition; **[CONJECTURE]** plausible but unproven; **[OPEN]** not settled by this suite.
Two governing rules of this report. **(R1)** every physical claim is tied to a named variable or invariant from the output; **(R2)** a zero or null output licenses only the *absence* of an effect (an identity confirmed, a bracket in involution, a traceless balance) — never a non-zero magnitude.

1. Raw Wolfram Data Anchors (the literal outputs)

Every row below is a verbatim kernel return. The left column is the tested Wolfram expression or invariant; the centre column is the literal printed value; the right column is the status the value supports. “ $\mathbf{1}_8$ ” denotes the 8×8 identity and “ z ” the 8×8 zero.

1.1 `representation.wl` — the defining Clifford relations

Wolfram expression / invariant	Literal output	Status
<code>AllTrue({e_i, e_j} - 2 $\eta_{ij} \mathbf{1}_8$ == z), all 36 pairs</code>	True	[EXACT]

This single Boolean certifies that the eight-dimensional real matrices generate Cl(3, 3) with signature $(+, +, +, -, -, -)$. It is the foundation of all 17 rows.

1.2 01_photon.centralizer.wl — the photon line is forced

Wolfram expression / invariant		Literal output	Status
$L_1^2 = L_2^2 = L_3^2 = -\mathbf{1}_8$	(temporal bivectors)	True	[EXACT]
$[L_1, L_2] = -2L_3$ and cyclic		{True, True, True}	[EXACT]
each $[L_i, \text{clock}] \neq z$	(clock = $e_1 + e_2 + e_3$)	{True, True, True}	[EXACT]
$\text{Solve}[(aL_1 + bL_2 + cL_3)\text{clock} = \text{clock}(\dots)]$		{{bb -> aa, cc -> aa}}	[EXACT]
$(L_1 + L_2 + L_3)^2 = -3\mathbf{1}_8$		True	[EXACT]
$J^2 = -\mathbf{1}_8$	($J = (L_1 + L_2 + L_3)/\sqrt{3}$)	True	[EXACT]
$\text{Tr}(J^2)$		-8	[EXACT]
$[J, W] = z$	(W the clock direction)	True	[EXACT]

The decisive anchor is the `Solve` return $\{\{\text{bb} \rightarrow \text{aa}, \text{cc} \rightarrow \text{aa}\}\}$: the only elements of $\mathfrak{su}(2)_T$ commuting with the clock satisfy $a = b = c$, a *one-dimensional* line.

1.3 02_charge.spectrum.wl — quantised charge

Wolfram expression / invariant		Literal output	Status
$\text{Sort}[\text{Eigenvalues}[J]]$		{-I, -I, -I, -I, I, I, I, I}	[EXACT]
$Q = -iJ$ Hermitian: $\text{ConjugateTranspose}[Q] - Q = z$		True	[EXACT]
$\text{Sort}[\text{Eigenvalues}[Q]]$		{-1, -1, -1, -1, 1, 1, 1, 1}	[EXACT]
charge after particle/antiparticle halving		$q = \pm \frac{1}{2}$	[DERIVED]

1.4 03_dispersion.wl — the relativistic mass shell

Wolfram expression / invariant		Literal output	Status
$G_a^2 = +\mathbf{1}_8, a = 1, 2, 3$	($G_a = W e_{3+a}$)	{True, True, True}	[EXACT]
$\{G_a, G_b\} = z$ for $a < b$		{True, True, True}	[EXACT]
$V^2 = +\mathbf{1}_8$	($V = e_4 e_5 e_6$, mass trivector)	True	[EXACT]
$\{G_a, V\} = z$		{True, True, True}	[EXACT]
$G_a^\top = G_a, V^\top = V$		{True, True}	[EXACT]
$H^\top = H$	($H = \sum_a G_a p_a + mV$)	True	[EXACT]
$H^2 - (p_1^2 + p_2^2 + p_3^2 + m^2)\mathbf{1}_8 = z$		True	[EXACT]

1.5 04_dirac_signature.wl — the Lorentzian signature

Wolfram expression / invariant	Literal output	Status
$\{\gamma^\mu, \gamma^\nu\} = 2 \text{diag}(+, -, -, -) \mathbf{1}_8$, all 16	True (grid all True)	[EXACT]
$(\gamma^0)^2 = +\mathbf{1}_8$	True	[EXACT]
$(\gamma^k)^2 = -\mathbf{1}_8$, $k = 1, 2, 3$	{True, True, True}	[EXACT]
$\text{Tr}(\gamma^0)$	0	[EXACT]
$\text{Sort}[\text{Eigenvalues}[\gamma^0]]$	{-1,-1,-1,-1,1,1,1,1}	[EXACT]

1.6 05_spin_curvature.wl — spin from boost curvature

Wolfram expression / invariant	Literal output	Status
$W^2 = +\mathbf{1}_8$, $J_3^2 = -\mathbf{1}_8$, $G_X^2 = +\mathbf{1}_8$	True; True; True	[EXACT]
$\{W, e_4\} = \{W, e_5\} = \{W, e_6\} = z$	{True, True, True}	[EXACT]
collapse identity $W e_4 W = -e_4$	True	[EXACT]
$[G_X, G_Y] = -2J_3 = -2e_4 e_5$	True	[EXACT]
$[G_Y, G_Z] = -2J_1$	True	[EXACT]
$[G_Z, G_X] = -2J_2$	True	[EXACT]
<i>supplementary verification (this report), same kernel:</i>		
$J_i^2 = -\mathbf{1}_8$ (spatial rotations $J_1 = e_5 e_6$, etc.)	{True, True, True}	[EXACT]
$[J_1, J_2] = +2J_3$ and cyclic	True	[EXACT]
$[G_X, G_Y] = +2J_3$?	False	[EXACT]

The last three rows isolate the sign that the original script does not print: rotations close with structure constant +2, boosts with -2 — opposite signs.

1.7 06_gravity_double_contraction.wl — the Einstein tensor

Wolfram expression / invariant	Literal output	Status
$n = 4$: Riemann-symmetry residual	0	[EXACT]
$n = 4$: (α, β) in $C = \alpha \text{Ric} + \beta \delta R$	{-4, 2}	[EXACT]
$n = 4$: max residual of $C - (\alpha \text{Ric} + \beta \delta R)$	0	[EXACT]
$n = 6$: Riemann-symmetry residual	0	[EXACT]
$n = 6$: (α, β)	{-4, 2}	[EXACT]
$n = 6$: max residual	0	[EXACT]

1.8 07_constraint_killing.wl — first class = Killing

Wolfram expression / invariant	Literal output	Status
$\{\phi, C\} + \frac{1}{2}(\mathcal{L}_u g)^{NK} P_N P_K$ (residual)	0	[EXACT]
$\{\phi_1, \phi_2\}$ on the flat surplus plane	0	[EXACT]

1.9 run_all.wl — consolidated verdict

The master runner reproduces all seventeen rows and returns a single Boolean.

Row	Identity	Result
1	$\{e_i, e_j\} = 2\eta_{ij}\mathbf{1}_8$	PASS
2	$W^2 = +\mathbf{1}_8$	PASS
3	$J^2 = -\mathbf{1}_8$	PASS
4	$\text{Tr}(J^2) = -8$	PASS
5	$[J, W] = z$	PASS
6	centralizer of W in $\mathfrak{su}(2)_T$ is one-dimensional $= J$	PASS
7	$G_a^2 = +\mathbf{1}_8, \{G_a, G_b\} = z$	PASS
8	$[G_X, G_Y] = -2e_4e_5$ and cyclic	PASS
9	$V^2 = +\mathbf{1}_8, \{G_a, V\} = z, V^\top = V$	PASS
10	$H = \sum_a G_a p_a + mV$ real symmetric	PASS
11	$H^2 = (p^2 + m^2)\mathbf{1}_8$	PASS
12	$\{\gamma^\mu, \gamma^\nu\} = 2 \text{diag}(+, -, -, -)$	PASS
13	$\text{Tr}(\gamma^0) = 0, \text{spec}(\gamma^0) = \{(+1)^{\times 4}, (-1)^{\times 4}\}$	PASS
14	$\text{spec}(J) = \{(+i)^{\times 4}, (-i)^{\times 4}\}$	PASS
15	$\delta_{drs}^{apq} R^s{}_{pq} = -4G^a{}_d \quad (n = 4, 6)$	PASS
16	$\{\phi_1, \phi_2\} = 0$ (flat surplus plane)	PASS
17	$\{\phi, C\} = -\frac{1}{2}(\mathcal{L}_u g)PP; = 0 \Leftrightarrow \text{Killing}$	PASS
ALL 17 ROWS PASS ?		True

2. Mathematical Symmetries (what the math allows)

The outputs of Section 1 are not eighteen unrelated facts; they are the fingerprints of a small set of algebraic structures. This section names each structure and shows, with worked micro-derivations, exactly which anchor forces it.

2.1 The algebra and its signature

Row 1 fixes the ambient object: a real associative algebra on six generators with $\{e_i, e_j\} = 2\eta_{ij}\mathbf{1}$, $\eta = \text{diag}(+, +, +, -, -, -)$, i.e. $\text{Cl}(3, 3)$, faithfully realised on $\mathbb{R}^8$. Everything downstream is an *internal* statement in this fixed representation; no further input is added.

2.2 One temporal $\mathfrak{su}(2)$ and a unique invariant direction

The three temporal bivectors L_a satisfy $L_a^2 = -\mathbf{1}$ and $[L_1, L_2] = -2L_3$ (cyclic), so they span $\mathfrak{su}(2)_T$. The clock $W \propto e_1 + e_2 + e_3$ commutes with no single L_a (row, all $\neq z$). The question “what in $\mathfrak{su}(2)_T$ commutes with W ?” is therefore non-trivial, and the kernel answers it exactly.

Worked example: the photon line is forced

Write $X = aL_1 + bL_2 + cL_3$ and impose $[X, \text{clock}] = 0$. The kernel returns $\{\{\mathbf{bb} \rightarrow \mathbf{aa}, \mathbf{cc} \rightarrow \mathbf{aa}\}\}$, i.e. $b = a, c = a$. The centraliser is the single line $\mathbb{R}(L_1 + L_2 + L_3)$. Normalising by

$(L_1 + L_2 + L_3)^2 = -31$ gives $J = (L_1 + L_2 + L_3)/\sqrt{3}$ with $J^2 = -\mathbf{1}$, $\text{Tr}(J^2) = -8$, $[J, W] = 0$. The unbroken gauge direction is *forced*, not chosen.

2.3 A Hermitian charge with an integer spectrum

Since $J^2 = -\mathbf{1}$, the operator $Q = -iJ$ is Hermitian ($Q^\dagger - Q = z$) and squares to $+\mathbf{1}$, so its eigenvalues lie in $\{+1, -1\}$. The kernel makes this concrete: $\text{spec}(J) = \{(+i)^{\times 4}, (-i)^{\times 4}\}$ and hence $\text{spec}(Q) = \{(+1)^{\times 4}, (-1)^{\times 4}\}$. The eight-real-dimensional module carries the two charge eigenspaces in equal 4+4 measure — a $U(1)_J$ acting with quantised charge.

2.4 An anticommuting symmetric quadruple and the mass shell

Rows 7–11 exhibit four real-symmetric matrices G_1, G_2, G_3, V that mutually anticommute and each square to $+\mathbf{1}$. That is precisely a Euclidean Clifford quadruple, and it forces the Dirac square.

Worked example: the Dirac square

With $H = G_1 p_1 + G_2 p_2 + G_3 p_3 + mV$,

$$H^2 = \sum_a G_a^2 p_a^2 + m^2 V^2 + \sum_{a < b} \{G_a, G_b\} p_a p_b + \sum_a \{G_a, V\} m p_a.$$

The anchors $G_a^2 = V^2 = +\mathbf{1}$ keep the diagonal; the anchors $\{G_a, G_b\} = z$ and $\{G_a, V\} = z$ delete *every* cross term. What remains is $H^2 = (p_1^2 + p_2^2 + p_3^2 + m^2)\mathbf{1}$, verified as a vanishing residual. Because H is real symmetric (row 10) its spectrum is real, giving $\text{spec}(H) = \pm\sqrt{p^2 + m^2}$ exactly. **[EXACT]** for the identity; **[DERIVED]** for the dispersion reading.

2.5 The signature is read off, not imposed

Setting $\gamma^0 = W$, $\gamma^k = e_{3+k}$, row 12 returns the full Clifford grid $\{\gamma^\mu, \gamma^\nu\} = 2 \text{diag}(+, -, -, -)$. The split $(\gamma^0)^2 = +\mathbf{1}$ versus $(\gamma^k)^2 = -\mathbf{1}$ is exactly mostly-minus Minkowski. Row 13 adds $\text{Tr}(\gamma^0) = 0$ with $\text{spec}(\gamma^0) = \{(+1)^{\times 4}, (-1)^{\times 4}\}$: a traceless time direction splitting the module into balanced positive- and negative-energy halves.

2.6 Boost curvature is rotation valued (the not-shown derivation)

Row 8 is the one substantive identity left unworked in the source text; the kernel supplies it.

Worked example: the collapse identity

Because $W^2 = +\mathbf{1}$ and $\{W, e_4\} = 0$,

$$We_4W = W(-We_4) = -W^2e_4 = -e_4 \quad (\text{kernel: } We_4W = -e_4, \text{ True}).$$

Hence $G_X G_Y = (We_4)(We_5) = (We_4W)e_5 = -e_4e_5$ and $G_Y G_X = -e_5e_4 = +e_4e_5$, so $[G_X, G_Y] = -2e_4e_5 = -2J_3$ (and cyclic), each verified as a vanishing residual.

The supplementary rows of §1.6 sharpen the reading: the spatial rotations themselves close as $[J_1, J_2] = +2J_3$, while the boosts produce $[G_X, G_Y] = -2J_3$. The relative minus sign is the algebraic signature of non-compactness,

$$[J_i, J_j] = +2 \epsilon_{ijk} J_k, \quad [K_i, K_j] = -2 \epsilon_{ijk} J_k,$$

the $\mathfrak{so}(3,1)$ pattern in which two boosts close onto a compact rotation. The curvature of the boost connection is therefore $\mathfrak{su}(2)_S$ -valued, with holonomy in $\text{Spin}(3) = \text{SU}(2)$.

2.7 The double-dual contraction is the Einstein tensor

On a generic tensor carrying the four Riemann symmetries (kernel: symmetry residual = 0), the rank-three generalised Kronecker contraction evaluates to

$$\delta_{drs}^{apq} R^{rs}_{pq} = \alpha R^a_d + \beta \delta^a_d R, \quad (\alpha, \beta) = (-4, 2),$$

with component residual 0 in *both* $n = 4$ and $n = 6$. Thus $\delta_{drs}^{apq} R^{rs}_{pq} = -4(R^a_d - \frac{1}{2}\delta^a_d R) = -4G^a_d$. The coefficient -4 is combinatorial: it does not depend on the dimension.

2.8 First-class closure equals the Killing condition

For a general inverse metric $g^{NK}(x)$ and vector field $u^M(x)$, with $\phi = u^M P_M$ and $C = \frac{1}{2}g^{NK}P_N P_K$, the symbolic Poisson bracket returns (residual 0)

$$\{\phi, C\} = -\frac{1}{2}(\mathcal{L}_u g)^{NK} P_N P_K,$$

so $\{\phi, C\} = 0 \Leftrightarrow \mathcal{L}_u g = 0$. Two constant surplus directions give $\{\phi_1, \phi_2\} = 0$: the surplus plane is flat and its flows are in involution.

3. Physical Interpretations & Boundary Limits (where the theory applies/-fails)

Each structure of Section 2 admits a physical reading. Under rule (R1) each reading is tagged with the anchor that carries it; under rule (R2) no zero output is inflated into a magnitude. The recurring boundary is the same throughout: this suite fixes *form*, never *scale*.

3.1 Electromagnetism and charge

Reading. The one-dimensional centraliser (anchor: $\text{Solve} \Rightarrow a=b=c, [J, W] = 0$) identifies a single unbroken $U(1)_J$ — the photon direction — as a forced consequence of demanding compatibility with the clock W . Its generator $Q = -iJ$ has integer spectrum ± 1 (anchor: $\text{spec}(Q) = \{(+1)^{\times 4}, (-1)^{\times 4}\}$), giving charge quantisation; the 4+4 split assigns $q = \pm \frac{1}{2}$ to a particle/antiparticle pair. **[DERIVED]**

Boundary limit

The spectrum is dimensionless: it fixes the *ratio* structure of charge, not the electromagnetic coupling. The output contains no number that can be read as e or α_{em} ; by (R2) the normalisation of physical electric charge is **[OPEN]**. The assignment $q = \pm \frac{1}{2}$ is a labelling of eigenspaces, not a measured quantum number.

3.2 The relativistic dispersion

Reading. The exact collapse $H^2 = (p^2 + m^2)\mathbf{1}$ (anchor: vanishing residual, plus $H^\top = H$) yields the on-shell relation $E = \pm \sqrt{p^2 + m^2}$. Crucially the mass term is the grade-three trivector $V = e_4 e_5 e_6$: the anchors $V^2 = +\mathbf{1}$ and $\{G_a, V\} = 0$ are what make the square relativistic. **[EXACT]** (identity) / **[DERIVED]** (dispersion).

Boundary limit

m enters H as a free symbol; no output assigns it a value, so the fermion mass *scale* is **[OPEN]**. The result is also conditional on the mass structure: relativistic covariance here is carried specifically by the grade-three V through $\{G_a, V\} = 0$. A different mass operator that fails this anticommutator would leave residual cross terms and a non-relativistic $E \propto p$ shell; enumerating the admissible Lorentz-covariant mass structures is **[OPEN]** and outside this suite.

3.3 Lorentzian signature and the Dirac content

Reading. Mostly-minus Minkowski is derived, not assumed: the grid $\{\gamma^\mu, \gamma^\nu\} = 2 \text{diag}(+, -, -, -)$ with $(\gamma^0)^2 = +1$, $(\gamma^k)^2 = -1$ fixes the local signature $(1, 3)$. The balance $\text{Tr}(\gamma^0) = 0$ with $\text{spec}(\gamma^0) = \{(+1)^{\times 4}, (-1)^{\times 4}\}$ encodes one Dirac fermion together with its conjugate. **[EXACT]** (signature) / **[DERIVED]** (particle content).

Boundary limit

By (R2) the zero trace says only that positive- and negative-energy sectors are *balanced* — no net chiral or energy asymmetry may be inferred from it. The statement is local (tangent-space and spinor signature); it does not by itself establish a globally Lorentzian dynamical spacetime, nor the number of generations.

3.4 Spin

Reading. The boost commutator is rotation valued, $[G_X, G_Y] = -2J_3$ (cyclic), with the opposite sign to $[J_1, J_2] = +2J_3$ (supplementary anchors). The boost field strength therefore has an $\mathfrak{su}(2)_S$ -valued quadratic part acting on the spinor as spin, with holonomy in $\text{Spin}(3) = \text{SU}(2)$ — the source of half-integer spin. **[EXACT]** (commutators) / **[DERIVED]** (spin reading).

Boundary limit

This identifies the *group* in which spin holonomy lives; it is not a derivation of the spin-statistics connection, nor of the spin content of any state beyond this representation. Those remain **[OPEN]**.

3.5 Gravity

Reading. The kinematic object of general relativity appears exactly: the double-dual contraction equals $-4G^a_d$ (anchor: $(\alpha, \beta) = (-4, 2)$ with residual 0 in $n = 4$ and $n = 6$). The Einstein tensor is thus an algebraic identity of the curvature, dimension-independent. **[EXACT]** (identity) / **[DERIVED]** (gravitational reading).

Boundary limit

The contraction is purely combinatorial: it contains no coupling. By (R2) nothing here fixes Newton's constant, and the identity alone is not a field equation — a variational principle and a matter source would be needed to promote G^a_d to dynamics. Ghost-freedom of the *full* $\text{Cl}(3, 3)$ lift (as opposed to the minimal sub-model) is not addressed by this suite and is **[OPEN]**.

3.6 Constraints and consistent reduction

Reading. First-class closure of the surplus constraints is *equivalent* to the Killing condition (anchor: $\{\phi, C\} + \frac{1}{2}(\mathcal{L}_u g)PP$ residual = 0), and two constant surplus directions are in involution (anchor: $\{\phi_1, \phi_2\} = 0$, a flat surplus plane). The reduction to the physical M^4 is therefore consistent exactly along Killing directions. **[EXACT]** (identity) / **[DERIVED]** (reduction reading).

Boundary limit

By (R2) the vanishing brackets certify involution — the *absence* of an anomaly along these directions — not a non-zero force. Closure is guaranteed only where u is Killing; global ghost-freedom across all strata, and the dynamics at the singular tip, lie beyond this symbolic check.

Scope verdict

All seventeen rows return **PASS** and the runner returns **True**. The algebraic skeleton of the Electromagnetic–Sphere construction — a forced photon line, quantised charge, an exact relativistic mass shell, an emergent Lorentzian signature, rotation-valued boost curvature (spin), the Einstein tensor as a curvature identity, and Killing-equivalent first-class closure — is established **exactly** in the real 8×8 $\text{Cl}(3,3)$ representation. No output in the suite fixes a *magnitude*: the electromagnetic coupling, the mass scale, Newton’s constant, the full- $\text{Cl}(3,3)$ ghost question, the colour sector, and the monopole/Witten effect are untouched here. **Verdict: structure complete, magnitudes open.**

Appendix — reproduction note

The kernel evaluations that produced every value above were: (i) the inlined $\text{Cl}(3,3)$ generators of `representation.wl` with $\eta = \text{diag}(+, +, +, -, -, -)$; (ii) Boolean identity tests of the form `Simplify[lhs - rhs] == 0` on 8×8 matrices, with `==` the structural-equality test against the zero matrix; (iii) `Eigenvalues/Sort` for the spectra of J , $Q = -iJ$ and γ^0 ; (iv) for gravity, projection of a random integer tensor (seeds 171 and 66) onto the algebraic-curvature subspace, followed by a two-point linear fit of the double-dual contraction against Ric and δR and a full-component residual; and (v) for the constraint sector, a canonical Poisson bracket $\{F, G\} = \sum_k (\partial_{x^k} F \partial_{P_k} G - \partial_{P_k} F \partial_{x^k} G)$ on a general x -dependent inverse metric. The messages `Symbol::undefined2` and `Solve::svars` appeared and are benign. The identity matrix is typeset **1** throughout (the blackboard-bold form is deliberately avoided to prevent any missing-glyph fallback).